Trusting the Invisible: A Usability Study Plan to Assess Trust in Real-World Deployments of Cryptographic Systems

TANMAYI JANDHYALA, University of Waterloo, Canada

Users tend to rely on institutional trust, especially when evaluating privacy guarantees advertised by service providers. They rarely interrogate cryptographic implementations as they can be technically difficult to comprehend. Instead, they defer to brand reputation, usability, or surface-level claims of security. This deference can lead to misplaced trust when services use implementations of outdated or weak cryptographic mechanisms but still present themselves as secure. While folk models and trust heuristics are well documented, little is known about whether this trust can be disrupted or reshaped through design interventions during usability testing. This study plan aims to examine how user trust in digital security systems, such as password managers, encrypted messaging apps, and two-factor authentication platforms, could be shaped more by institutional branding and interface familiarity than by actual security features that the underlying cryptographic mechanisms offer. This falls under the umbrella of how cryptographic systems are deployed in the real-world in terms of design and user experience with the applications. The broader goal is to evaluate whether trust in well-known institutions overrides users' concern for verifiable cryptographic guarantees, and whether usability interventions can help close this gap - but it goes beyond the scope of this study plan. Findings could inform the design of more transparent and trustworthy user-facing systems, especially for emerging cryptographic deployments such as post-quantum systems which offer strong technical protections but further lack mainstream familiarity than existing systems.

Additional Key Words and Phrases: Trust, Cryptography, Usability

1 Introduction and Related Work

Prior research on user trust in digital systems reveals that trust is shaped not only by technical assurances but also by social cues, mental models, and infrastructural expectations. Riegelsberger et al [10] emphasize that human trust is often affective and interpersonal, and that users respond to visual cues, such as photographs or signs of social presence, when assessing the trustworthiness of online systems, even when these cues are inconsistent or unrelated to the actual security properties of the system. Their work shows that such design elements can impact not just perceived trust, but also a user's ability to correctly identify trustworthy services. For users to "trust" a party that implements a specific piece of cryptography as its primary offering of service, service providers would have to go beyond merely deploying code that does a specific functionality without informing the users of their true benefits (in a way that is adequately comprehensible within their limited interaction with the service). This problem becomes more acute when law enforcement access or regulatory compliance creates additional pressure for companies to retain access to user data, further complicating the security landscape. In such contexts, trust is not just a usability or perceptual issue, it becomes a matter of liability. If institutional actors deploy weak or obsolete cryptographic systems while continuing to signal strong security (e.g., through padlock icons, branding, or vague encryption claims), then users may unwittingly face the legal and financial consequences of breaches they had no way of anticipating or understanding.

McDonald and Cranor [8] explore how users make privacy decisions in the context of behavioral advertising. They show that users often misunderstand how cookies, tracking, and data use function, relying instead on superficial cues like the presence of a privacy policy. Even when privacy-enhancing features are available, users may not know how to use them effectively or may assume that institutional actors are automatically complying with ethical norms. Although this study

Author's Contact Information: Tanmayi Jandhyala, tjandhyala@uwaterloo.ca, University of Waterloo, Waterloo, ON, Canada.

pertains to deployments of cryptosystems, the philosophy of it could apply to a discrepancy between the security advertised and the security implemented.

Wash [12] investigated how non-expert users develop folk models of computer security threats. These simplified conceptualizations guide user decision-making in ways that often conflict with expert advice. Users frequently ignore technical guidance when it clashes with their lived experience or intuitive understanding of threats. More broadly, Bruun et al [4] position trust and distrust as infrastructural forces in the design and deployment of cryptographic technologies. They contrast traditional institutional trust with the idea of "trustless trust" in cryptographic systems like blockchains or secure multi-party computation. However, they caution that even in trustless systems, political accountability and user comprehension remain pressing concerns. Numerous studies have shown that users' adoption of security technologies is deeply influenced by their perceptions of trust. Alkaldi and Renaud [2] conducted interviews exploring why individuals choose to adopt or reject smartphone password managers. A recurring theme in participant responses was the importance of perceived trustworthiness of the service provider. As one participant noted, "I don't know who's behind the app, so I wouldn't trust it with my passwords," illustrating how unfamiliar branding and lack of transparency about what the underlying systems does to secure user data can significantly erode user confidence, even when technical protections exist. Other users expressed concerns about how password managers store sensitive data, particularly if they felt they had "no control" or lacked understanding of how encryption worked. Crucially, the study reveals a persistent disconnect between cryptographic security and user trust: even when technically secure systems are in place, users may not feel secure unless the system signals trustworthiness in familiar or interpretable ways.

These works illustrate that trust is a multidimensional phenomenon, shaped by social, technical, and institutional factors. While there is a need to separate the way cryptosystems are deployed and what users are made to know about it, the study is motivated by the fact that not all details of security or privacy-preservation need to be encapsulated. Further, the manner in which they are presented to the user is also essential in building their feelings of trustworthiness and to ensure their longevity of use of the product. This also highlights an interesting gap: while users often rely on institutions and interface cues to guide their trust decisions, little is known about how and whether those trust decisions might be reshaped when users are made aware of cryptographic guarantees on an implementation-level. If developers, architects, and policy-makers can collaboratively fix such usability gaps identified via studies like this (or better ones), it could be possible to create slightly safer user experiences.

2 Research Questions

This study aims to investigate how users evaluate the trustworthiness of cryptographic systems in real-world digital services, with a focus on mental models, brand heuristics, and perceived risk. The overarching research question is:

> **RQ1:** How do users form trust judgments about digital services that deploy cryptographic protections, and to what extent are those judgments shaped by institutional familiarity, visual cues, and perceived professionalism?

To operationalize this investigation, we explore the following sub-questions:

RQ1.1: How do users interpret security claims made by services offering password management, encrypted email messaging, or multi-factor authentication?

RQ1.2: What role do usability factors (such as interface design, terminology, transparency cues) play in shaping perceived security?

RQ1.3: Do discrepancies between cryptographic transparency and institutional branding affect users' trust decisions?

3 Planned Methodology

3.1 Overview

The study uses a qualitative research design, drawing methodological inspiration primarily from McDonald's investigation of users' perceptions in online advertising [8] and Riegelsberger et al.'s trust frameworks [10]. Additionally, it adapts validated measures from McKnight et al.'s trust in technology scales [9]. The study proceeds in two stages: first, by qualitatively exploring users' existing mental models and beliefs about cryptographic security and institutional trust; second, by using these insights to inform a structured survey intended to capture trust perceptions.

3.2 Phase 1: Qualitative Exploration [1]

Although the initial plan lead to two phases of carrying out the study, this research plan only focuses on the first phase so as to draw attention better to related work and the inferences that can be made by potential results of the structured surveys. The primary objective is to uncover and understand the mental models that users hold about institutional trust and cryptographic security in digital systems. The aim is to design a study that offers developers and cryptography engineers, insights into the following aspects of users' perceptions of trust in cryptographic systems:

(1) **Mental Models of Trust:** How participants conceptualize their beliefs about what makes a service "secure."
(2) **Understanding of Cryptographic Concepts:** What participants know (or assume) about encryption, digital signatures, and data privacy mechanisms, and whether they can recognize or explain these in everyday services.
(3) **Brand-Based Trust Heuristics:** The extent to which participants trust well-known service providers (regardless of the underlying cryptographic protocol), and the rationales they provide for this trust.
(4) **Perceived Risk and Threat Models:** Who participants believe might access their data, and what protection strategies (if any) they believe cryptographic mechanisms offer against these threats.

This study, as a stretch goal, also aims to explore the below user perception. The methodologies or the appendix has insufficient information to support this aspect of the study for the current version of this document.

* **Security Disclosure Interpretation:** How participants interpret surface-level security disclosures (e.g., "end-to-end encryption," "this payment gateway is encrypted for the protection of your data"), and whether they influence trust judgments.

For this, our primary goal is to conduct semi-structured interviews, following a modified version of the mental models methodology [6, 8], by selecting approximately 15-20 participants. For the sake of writing the entire plan, the results from these initial interviews are assumed. The recruitment would have aimed for demographic diversity to capture a broad range of perspectives and to achieve thematic saturation.

3.2.1 Use of Focus Groups. Designing semi-structured interviews requires meticulous designing of interview guidelines to effectively record the mental models and perceived understanding of users. Dechand et al. [6] use focus groups to develop a final interview guideline. This involves a single

[1] Phase 2 is out-of-scope of this research plan, but is something I hope to better comprehend and explore one day as a part of my research.

skilled researcher to lead and moderate focus group discussions for creating an initial guideline and building on top of it based on the responses from the participants in the focus group. A few iterations of these groups are conducted so as to improve on the guidelines each time. For this study, this method seemed like an efficient (cost- and time-effective) way to be prepared with initial guidelines for the structured surveys.

The focus group would typically be used to define participants' mental models of trust, and in shaping questions to measure user perceptions of cryptosystems more clearly. A small number of initial focus group sessions (approximately 2 sessions with 4-6 participants each) will be conducted primarily to inform the development of a comprehensive and effective interview script. Participants will be compensated with refreshments. Focus groups will gather initial insights into user understanding, perceptions, and language around cryptographic systems and trust in digital services. Participants will discuss openly how they perceive concepts such as encryption, authentication, and trust indicators. Based on these discussions, a refined and more structured interview script will be produced (see Appendix C C).

3.2.2 Individual Interviews. Following the focus group stage, we will conduct detailed, semi-structured individual interviews with 8–10 participants to explore participants' general perceptions of trust, security, and usability in services that claim to use some kind of cryptography. Rather than evaluating specific interfaces at this stage, these interviews will be designed to elicit users' mental models, trust heuristics, and perceived risks and benefits associated with cryptographic protections in everyday digital tools (e.g., password managers, 2FA systems, and encrypted communication platforms [2]). Participants will not be shown mock-ups or screenshots during this phase. Instead, open-ended prompts will guide discussions around what cues, labels, or assurances users associate with digital safety, how they interpret terms like "encryption" or "secure login," and what makes them feel a service is trustworthy. These insights will help identify common themes and surface design-relevant heuristics that users rely on when evaluating the security of a system.

Findings from this stage will directly inform the design of the follow-up mock-up stimuli for the survey. For example, frequently mentioned cues such as padlock icons, brand reputation, or visible statements of encryption will be incorporated into the visual mockups used in the survey phase. This approach ensures that the quantitative survey measures are grounded in real user language and trust reasoning patterns identified during the qualitative interviews.

As Riegelsberger et al. [11] argue, trust emerges not only from contextual signals but also from intrinsic, internalized norms shaped through interaction and understanding. If digital environments rely solely on external assurance mechanisms such as branding or technical guarantees without supporting users in forming coherent mental models, trust may degrade into mere reliance. This distinction is critical: reliance can be brittle and inflexible, often failing when contextual cues break down. Inspired by Riegelsberger et al. [11], the interviews will target the following trust dimensions. These terms were directly taken from their paper and are not my own:

(1) *Perceived professionalism:* Do users associate smooth design and lack of technical errors with stronger security? How much do visual cues and branding influence their comfort?
(2) *Temporal embeddedness:* Do signs of long-term investment (e.g., brand familiarity, existing client base, loyalty incentives) increase perceived trust, even when security mechanisms are not made visible?

[2]These three services were selected because they represent common categories of cryptographic applications that are likely to expose users to a broad range of potential security cues. This list is not exhaustive, and I acknowledge the inherently subjective nature of this study.

(3) *Social embeddedness:* How do recommendations from friends, online reviews, or media shape their decisions to trust one service over another? (assessing this is beyond the scope of this study plan)
(4) *Understanding of cryptographic protections:* What do users understand or perceive about encryption, key protection, or transmission guarantees, and where do misunderstandings arise?

The last piece of the above will further help in capturing *Reliance vs. trust:* Are users merely relying on brand assurance, or do they form trust based on perceived internal properties of the system? Once it is clear what users perceive about cryptographic systems for their priacy or security, it becomes convenient to farm survey questions that also touch on this. These interviews will provide qualitative data to analyze user beliefs, heuristics, and potential misconceptions, grounding our findings in practical interactions aligned with previous research [8].

3.2.3 Assumptions about Focus Group and Interview Revelations: Since this research plan is forward-looking and neither focus groups nor individual interviews have yet been conducted, certain assumptions are made to ensure methodological consistency. For initial focus groups, we assume participants will generally express skepticism or uncertainty regarding the security of commonly-used digital services, despite continuing to actively use them. Participants may believe that services, including password managers, encrypted email or messaging apps, and two-factor authentication (2FA), can potentially be compromised by motivated individuals or companies with sufficient expertise or resources. However, many will still choose convenience over perceived risks, relying heavily on brand familiarity, visual cues, and ease-of-use rather than on actual technical assurances. Additionally, we anticipate that few participants will have a clear understanding of cryptographic concepts such as public-key encryption, instead simplifying security to passwords or symmetric encryption. As such, seeing a lot of security cues might overwhelm users in a way that they might not want to use the service. Insights gained from these anticipated revelations will guide the use of well-established, semi-structured survey questions, and in designing mock-ups that do not negatively control the result. emphasizing brand trust, visual indicators, and scenario-based discussions rather than technical cryptographic mechanisms.

3.2.4 Designing the Mock-ups. Appendix D shows two mock-ups designed to accompany the survey. These are intended to simulate realistic user interfaces that resemble existing applications for password management. Due to the limited time I had to complete this report, mock-ups for two-factor authentication and encrypted communication were not provided in this document. The plan for the study is for each mock-up to be constructed to manipulate visual cues related to trust and security without introducing confounding usability differences. The qualitative codes derived from individual interviews would have informed the mock-up design. For example, recurring mentions of interface professionalism and branding would be encoded as `interface_trust` and `brand_trust`, respectively. Because the results from the interviews were assumed, for this study plan, designs were based on those assumptions. Critically, we contrast two visual conditions across applications:

(1) **No Explicit Cryptographic Cues** [3]: These mock-ups show the logo and design of moderately popular [4] service providers with polished, professional user interfaces but minimal

[3] when we say cryptographic cues, it is in a way that non-technical users would find not-too-difficult to comprehend. This is incredibly hard to do, I have realized through this project.
[4] I chose moderately popular branding because highly popular ones might get biased responses due to their reputation in the media, news, and so on. I am not entirely sure if this reasoning is sound.

to no visual indicators of encryption or cryptographic security. They reflect real-world conditions where institutional trust is implicitly relied upon.

(2) **Clear Cryptographic Cues**: These mock-ups present fictional or minimally branded services, designed with prominent cryptographic indicators (e.g., lock icons, "end-to-end encrypted" labels, zero-knowledge claims). These simulate conditions where security is explicitly communicated through the interface rather than inferred through brand recognition.

Participants will view and rate each mock-up independently during the survey, and item order will be randomized to reduce priming or ordering effects.

3.2.5 Survey. Following the individual interviews, a broader online survey will quantify findings using validated scales adapted from McKnight et al. (Appendix A). Participants will be provided with mock-up images - one without obvious cues indicating cryptographic guarantees, and one with. The order of presenting them in the online survey would be: Image(s) with cues > Survey questions > Image(s) without cues > Same survey questions. OR: Image(s) without cues > Survey questions > Image(s) with cues > Same survey questions. Given the use of image mock-ups (that are on-interactive), a formal System Usability Scale (SUS) [3] assessment was considered irrelevant for the purposes of this study. Instead, usability is addressed qualitatively through the interviews and quantitatively through the brand-based trust and usability section of the survey. McKnight et al.'s validated trust-in-technology scales [9] will be used in this research study to construct structured questions given in the appendix. These scales measure users' perceptions of competence, reliability, and predictability of the technological aspects explicitly, providing an additional layer of construct validity. The survey thus ensures quantifiable data collection, complementing qualitative insights from the focus groups and individual interviews.

Participants for the survey will be recruited through university mailing lists, community forums, and social media platforms, supporting a range of technical and possibly non-technical backgrounds. Eligibility will be restricted to individuals over the age of 18, and who have used at least one of the following services: a password manager (could be a part of their browser too), an email messaging application, or a 2FA app in the past year. Participants will be asked to provide informed consent and will be offered a small honorarium for completion. Recruitment messaging will avoid security-specific language to reduce priming effects. The survey will be administered online using a secure platform. Participants will first complete a short demographics section, followed by four loosely-structured blocks:

(1) **Brand-based Trust Heuristics (Appendix A, Part 3, Part 4):** Participants will rate perceived trustworthiness based on real and fictional brand names, presented alongside generic descriptions of cryptographic services.
(2) **Perceived Risk and Threat Awareness (Appendix A, Part 2):** Participants will assess their perceptions of risk through brief vignettes describing scenarios involving familiar and unfamiliar cryptographic services.
(3) **Trust in Service Provider (Appendix A, Part 1, Part 4):** Participants will evaluate the perceived competence, reliability, and integrity of the service providers described.
(4) **Trust in Cryptographic Mechanism (Appendix A, Part 3, Part 5, Part 6):** Participants will rate their understanding and trust in the explicit cryptographic claims made by the services, assessing confidence in deployment practices of cryptographic systems.

The entire survey will take approximately 15–20 minutes to complete. All responses will be anonymized. Appendix **??** contains the survey questions.

3.2.6 Planned Evaluations. Quantitative data from survey responses will be analyzed primarily using descriptive statistics (means, standard deviations, medians) and inferential statistical techniques. Specifically, **paired t-tests** will compare differences in survey participants' perceived trust and risk across familiar versus unfamiliar brands.

3.2.7 Analysis. Additional exploratory analyses may be conducted based on other statistical techniques discussed in the course (such as one-way ANOVA for comparing means across multiple conditions, or correlation analyses to examine the relationship between trust scale scores and participant demographics or attitudes). These analyses will be determined by the characteristics of the data and the research questions that arise during interpretation. Although the primary contribution of this study aims to be qualitative, the structured and theory-informed quantitative component based on adapted trust scales (Appendix A) would help establish for a stronger internal validity across methods.

3.2.8 Limitations. Despite its multi-method approach, this study presents several limitations. First, although scenario-based interviews and online surveys aim to capture realistic decision-making processes, they rely on self-reported perceptions rather than observed behaviors. This limits our ability to assess how participants would act under stakes involving their own sensitive data or financial assets.

Further, participants may interpret cryptographic cues and security terminology differently based on prior experience, digital literacy, or cultural assumptions. This may lead to inconsistencies in responses, especially in qualitative interviews. To mitigate this, we have minimized jargon and included clarifying prompts, but residual ambiguity may still affect the internal validity of thematic analysis.

Trust ratings, specifically those collected via quantitative scales, may be disproportionately influenced by brand familiarity or user interface polish rather than accurate comprehension of the deployment of cryptographic guarantees. For example, authentication is rarely about encryption, however, we could not use the term "digital signature" or "authentication" itself because they might be confusing terms. We instead rely on the general understanding of encryption that users typically would have to measure their trust on systems that go beyond just encryption. Also, users might not realize while answering these questions that a whole cryptosystem sits beneath the application that they're using and so might misplace trust. While the study includes brand manipulation checks, these cannot fully eliminate the effect of well-established reputational heuristics. This may obscure whether participants are trusting the cryptographic system or the institutional identity behind it. It is mostly the latter, which is why this study further needs a lab study for usability where their interactions with these systems and trust perceptions are recorded in real-time through think-alouds or reflection questions.

I also do recognize that perceived risk judgments may be shaped by transient sociopolitical or media narratives, such as news coverage of data breaches or high-profile privacy failures. Finally, the study's findings may not generalize to populations with limited access to or familiarity with digital infrastructure, such as older adults, low-income users, or users in non-Western regulatory environments. Since the focus group and survey recruitment is likely to be skewed toward digitally literate individuals, the study primarily reflects mental models of moderate to high-frequency technology users.

References

[1] 1Password. 2025. 1Password Sign In. https://my.1password.com/signin?l=en. Accessed: 2025-07-29.

[2] Nora Alkaldi and Karen Renaud. 2016. Why do people adopt, or reject, smartphone password managers?. In *1st European Workshop on Usable Security-EuroUSEC 2016.*

[3] John Brooke et al. 1996. SUS-A quick and dirty usability scale. *Usability evaluation in industry* 189, 194 (1996), 4–7.

[4] Maja Hojer Bruun, Astrid Oberborbeck Andersen, and Adrienne Mannov. 2020. Infrastructures of trust and distrust: The politics and ethics of emerging cryptographic technologies. *Anthropology Today* 36, 2 (2020), 13–17.

[5] Canva. 2025. Webpage Templates. https://www.canva.com/s/templates?query=webpage. Accessed: 2025-07-29.

[6] Sergej Dechand, Alena Naiakshina, Anastasia Danilova, and Matthew Smith. 2019. In encryption we don't trust: The effect of end-to-end encryption to the masses on user perception. In *2019 IEEE European Symposium on Security and Privacy (EuroS&P)*. IEEE, 401–415.

[7] Karola Marky, Kirill Ragozin, George Chernyshov, Andrii Matviienko, Martin Schmitz, Max Mühlhäuser, Chloe Eghtebas, and Kai Kunze. 2022. "Nah, it's just annoying!" A deep dive into user perceptions of two-factor authentication. *ACM transactions on computer-human interaction* 29, 5 (2022), 1–32.

[8] Aleecia McDonald and Lorrie Faith Cranor. 2010. Beliefs and behaviors: Internet users' understanding of behavioral advertising. Tprc.

[9] D Harrison Mcknight, Michelle Carter, Jason Bennett Thatcher, and Paul F Clay. 2011. Trust in a specific technology: An investigation of its components and measures. *ACM Transactions on management information systems (TMIS)* 2, 2 (2011), 1–25.

[10] Jens Riegelsberger and M Angela Sasse. 2001. Trustbuilders and trustbusters: The role of trust cues in interfaces to e-commerce applications. In *Towards the E-Society: E-commerce, E-business, and E-government*. Springer, 17–30.

[11] Jens Riegelsberger, M Angela Sasse, and John D McCarthy. 2005. The mechanics of trust: A framework for research and design. *International journal of human-computer studies* 62, 3 (2005), 381–422.

[12] Rick Wash. 2010. Folk models of home computer security. In *Proceedings of the Sixth Symposium on Usable Privacy and Security* (Redmond, Washington, USA) *(SOUPS '10)*. Association for Computing Machinery, New York, NY, USA, Article 11, 16 pages. doi:10.1145/1837110.1837125

A Trust in Cryptographic Systems (Adapted from McKnight et al. [9] [5])

This appendix includes adapted items from McKnight et al.'s validated trust-in-technology scales [9], reworded to apply to real-world deployments of cryptographic systems. All items will be rated on a 7-point Likert scale ranging from "strongly disagree" (1) to "strongly agree" (7).

Part 1: Trusting Beliefs — Functionality

(1) This system seems to have the functionality I need.
(2) This system seems to have the features required to protect my data.
(3) This system seems to have the ability to secure what I want it to secure.

Part 2 Trusting Beliefs — Helpfulness

(4) This system seems to help me understand how my data is protected.
(5) This system seems to provide competent guidance (as needed) through help or support features.
(6) This system seems to provide the help I need to stay secure.
(7) This system seems to give very sensible and effective security advice when needed.

Part 3: Situational Normality — Cryptographic Interfaces

(8) I am totally comfortable using encryption-backed services.
(9) I always feel confident that the right things will happen when I use these services.
(10) It appears that things will be fine when I use systems like password managers or secure email.

Part 4: Structural Assurance — Trust Infrastructure

(11) I would feel safe using these services because they are backed by companies I can trust.
(12) Guarantees from companies make me feel okay using secure technologies.
(13) Legal protections and privacy policies help me feel safe using these systems.
(14) Having the backing of compliance standards or privacy laws makes me feel secure using tools that keep my data secure and protect my privacy online.

Part 5: General Faith in Cryptographic Technologies

(15) I believe that systems like this have the features needed to protect user data.
(16) I think technologies like this allow me to keep my information secure.

Part 6: Trusting Stance — Toward New Cryptographic Technologies

(17) My typical approach is to trust a new secure service until it proves untrustworthy.
(18) I usually trust cryptographic technologies until they give me a reason not to.
(19) I generally give the benefit of the doubt to new privacy-protecting systems.

B Codebook Used to Analyze Interview Responses (based on McKnight et al. [9] and Marky et al [7].'s work)

[5] I tried to keep this "adaptation" identical to the scale given in the paper, but that trust scale was for one service and because this study plan does not use an existing service as such, it needed to be adapted in my opinion.

Category	Code	Description
Mental Models of Encryption	encryption_surface	Basic awareness of encryption, but no technical understanding
	encryption_functional	Belief that encryption protects data from third parties (e.g., cloud provider)
	encryption_misconception	Incorrect belief, e.g., that a password alone equals encryption or that encrypted data is “invisible”
	authentication_confusion	Confusion between encryption and authentication
Trust Heuristics	brand_trust	Trust based on company brand, reputation, or familiarity
	interface_trust	Perceived trustworthiness based on professional UI, design, or usability
	feature_trust	Trust in explicit features like “end-to-end encryption” or “Two factor authentication”
	thirdparty_trust	Trust based on endorsements or use by peers, media, or institutions
Perceived Risks	password_visibility	Concern about the password manager accessing passwords
	data_leak_fear	Fear that password managers could leak passwords or they be “hacked"
	misuse_concern	Concern about third-party use of passwords or sensitive data (e.g., for advertising, surveillance)
Threat Models	provider_threat	Belief that the provider may be a potential attacker
	external_hacker	Concern about outsiders or hackers breaching the system
	insider_threat	Concern that employees or insiders of the provider can access files/emails/passwords
	accidental_exposure	Concern about password misplacement, wrong authentication, or accidental sharing
Usability Factors	ease_of_use	Perceived ease of navigating and using the platform
	transparency	Clarity of the system’s explanations and security claims
	learnability	Whether user feels they can learn to use security features without expert help
Behavioral Outcomes	file_avoidance	Avoidance of uploading sensitive files altogether
	risky_sharing	Sharing data despite risk, due to convenience or lack of alternatives
	security_feature_toggle	Turning off security features due to usability barriers

C Interview Script

The following script would be used to guide the interviews. Questions are designed to elicit participants' *mental models of encryption and authentication*, as well as their *trust heuristics*, *perceived risks and threat models*, and perceptions of *usability and brand-based trust*—particularly in the context of password management and file-sharing services that claim to use encryption.

(1) **Warm-up and Context Setting**
- Can you describe a time when you asked your browser or a dedicated app to remember your password?
- What made you feel confident or uneasy about doing so?

(2) **Understanding Cryptographic Architecture and Threat Models**
- How do you think this service protects your passwords from unauthorized access?
- Do you believe that the company remembering your passwords can read your passwords? Why or why not?
- When your password is auto-filled on a website, do you think it is still protected? Why?

(3) **Mental Models of Encryption and Authentication**
- Have you heard the term "encryption"? What does it mean to you in your own words?
- When a service says "your email is encrypted," what do you imagine is happening?
- What role do you think passwords or multi-step login processes play in protecting your data?

(4) **Trust Heuristics and Perceived Risk**
- Do you use login systems that ask for a second action, like clicking a button on another device?
- What makes you feel a system is trustworthy or not?
- Do you trust some brands or services more than others to keep your login information safe? Why?
- If something went wrong (e.g., unauthorized login or denied access), who would you hold responsible?

(5) **Brand-Based Trust and Usability Factors**
- Does the brand or reputation of a company affect whether you use their security features or services?
- How does the ease-of-use or design of a service influence your perception of its security?
- Do you associate polished design or intuitive features with stronger protection?

(6) **Closing**
- Is there anything else you want to share about how you think about encryption, secure payments, passwords, or trust in the digital services that you use everyday?
- What could companies do to make their security claims easier to understand or verify?

D Mock-ups

Currently, only one pair of mock-ups (for the password manager) have been made as an example. Please note that even though 1Password [1] is referenced, the mock-up does not exactly match the website sign-in page. Both the images were made using Canva [5], so Figure 2 is not an actual comparison to the how the existing tool itself performs to an non-existent tool.

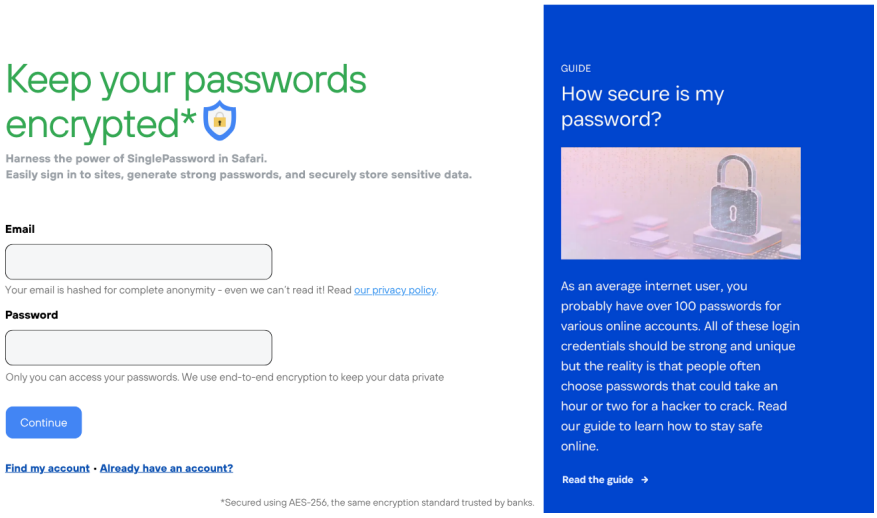

Fig. 1. Mock-up of a webpage with cryptography cues

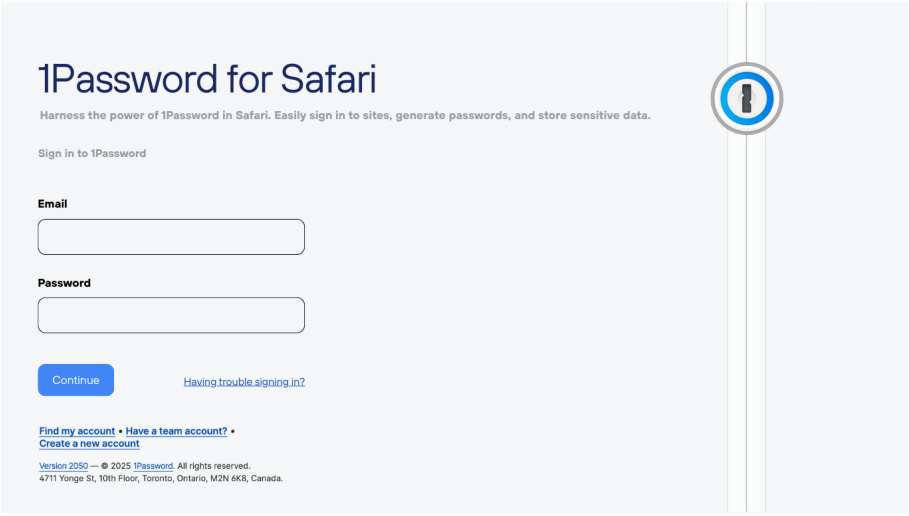

Fig. 2. Mock-up of a webpage of a moderately popular brand, but without cryptography cues